2017 NYJC JC2 Prelim 9758/1 Solution

<table>
<tr><th>Qn</th><th></th><th>Remarks</th></tr>
<tr><td>1</td><td>Sum of numbers in kth row = $\sum_{r=1}^{k} r = \frac{1}{2}k(k+1)$
Required sum = $\sum_{k=1}^{n} \frac{k(k+1)}{2}$
$= \frac{1}{2}\sum_{k=1}^{n}(k^2+k)$
$= \frac{1}{12}n(n+1)(2n+1)+\frac{1}{4}n(n+1)$
$= \frac{1}{12}n(n+1)(2n+1+3)$
$= \frac{1}{6}n(n+1)(n+2)$</td><td>A handful of students wrote $\sum_{r=1}^{k} k = \frac{1}{2}k(k+1)$, without realising that k is in fact a constant, and the expression is incorrect.</td></tr>
<tr><td>2(i)</td><td>Differentiating $2x-y^2=(x+y)^2$ ______(1)
implicitly with respect to x,
$2-2y\frac{\mathrm{d}y}{\mathrm{d}x}=2(x+y)\left(1+\frac{\mathrm{d}y}{\mathrm{d}x}\right)$
Where tangent is parallel to the x-axis, $\frac{\mathrm{d}y}{\mathrm{d}x}=0$.
$2=2(x+y)$
$y=1-x$ ______(2)
Sub (2) in (1),
$2x-(1-x)^2=(x+1-x)^2$
$x^2-4x+2=0$
$x=\frac{-(-4)\pm\sqrt{(-4)^2-4(2)}}{2}=2\pm\sqrt{2}$
When $x=2-\sqrt{2}$, $y=1-\left(2-\sqrt{2}\right)=-1+\sqrt{2}$
When $x=2+\sqrt{2}$, $y=1-\left(2+\sqrt{2}\right)=-1-\sqrt{2}$</td><td>Many students stopped at equation (2) and incorrectly concluded that it was the required equation of tangent.</td></tr>
</table>

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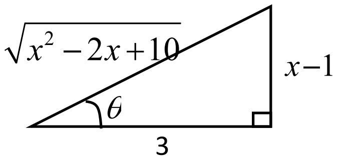
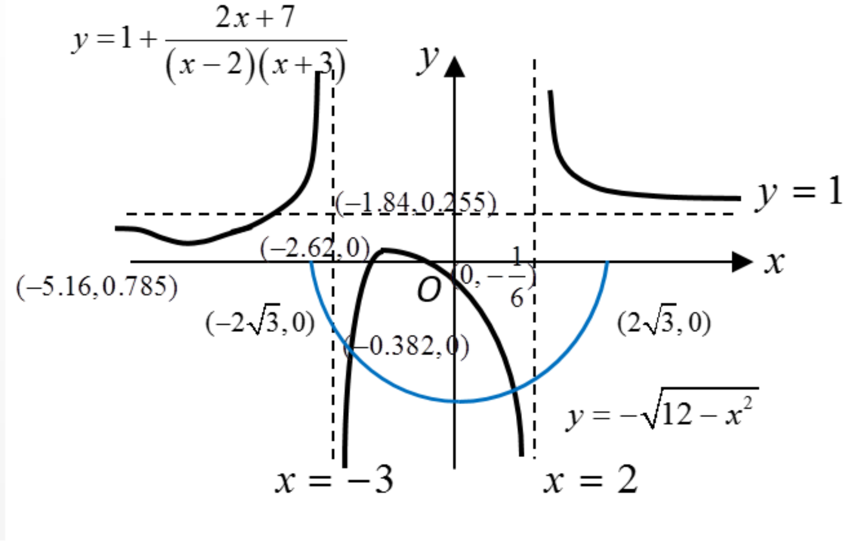
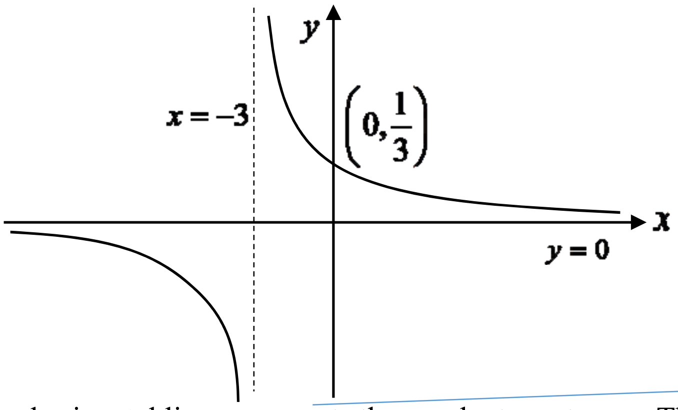
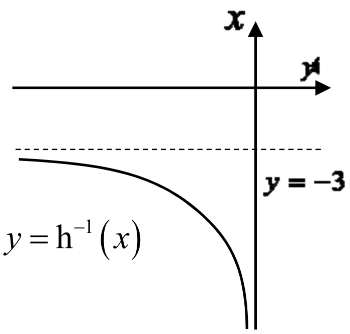
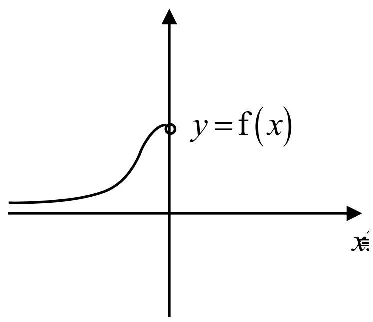
Qn		Remarks
2(ii)	$2-2y\frac{\mathrm{d}y}{\mathrm{d}x}=2(x+y)\left(1+\frac{\mathrm{d}y}{\mathrm{d}x}\right)$ $2=2(x+y)+2(x+2y)\frac{\mathrm{d}y}{\mathrm{d}x}$ $2=2(x+y)+2(x+2y)\frac{\mathrm{d}y}{\mathrm{d}x}$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1-(x+y)}{x+2y}$ When $x=0,\ y=0$, $-\frac{1}{\frac{\mathrm{d}y}{\mathrm{d}x}}=0$. Hence normal to C at the origin is $y=0$. When $x=2,\ y=-2$, $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{-2}$ Tangent to C at $A(2,-2)$, $y-(-2)=-\frac{1}{2}(x-2)$ Where the normal and the tangent intersect, $2=-\frac{1}{2}(x-2)$ $x=-2$ Area of triangle $OAB=\frac{1}{2}(2)(2)=2\text{ units}^2$	This part was badly done as many wrote the equation of normal to C as $x=0$, instead of $y=0$. Due to this error, they were unable to obtain the required area.
3(i)	Since a is real, the polynomial equation has real coefficients, and thus all non-real roots must be in conjugate pairs. Since the degree of the polynomial is three, there will be 3 roots. The highest even number below 3 is 2.	Candidates showed vague understanding of conjugate root theorem. Note that it does not imply that there WILL be complex conjugate roots.
3(ii)	$z^3+az^2+az+7=0$ $(-7)^3+a(-7)^2+a(-7)+7=0$ $a=8$ $z^3+8z^2+8z+7=0$ $(z+7)(z^2+z+1)=0$ $z=-7$ or $z=\frac{-1\pm\mathrm{i}\sqrt{3}}{2}$ $z=7e^{i\pi},\ \mathrm{e}^{\frac{i2\pi}{3}},\mathrm{e}^{-\frac{i2\pi}{3}}$	If candidates sub in $x=a+bi$ and $x=a-bi$ and proceed to solve for a and b, they will be penalised as the complex conjugate roots may not even exist in the first place.

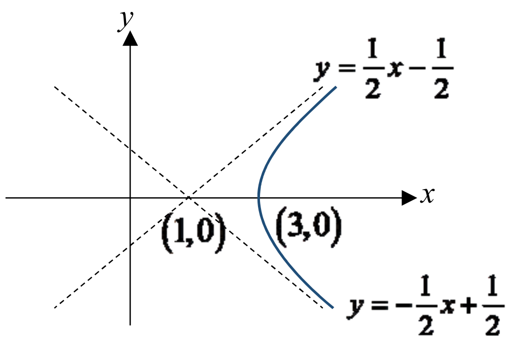
Qn		Remarks
3(iii)	$-\mathrm{i}z^3-8z^2+8\mathrm{i}z+7=0$ $(\mathrm{i}z)^3+8(\mathrm{i}z)^2+8(\mathrm{i}z)+7=0$ From (ii), replace z with $\mathrm{i}z$ $\mathrm{i}z=7e^{i\pi},\ \mathrm{e}^{\frac{i2\pi}{3}},\mathrm{e}^{-\frac{i2\pi}{3}}$ $\Rightarrow z=-\mathrm{i}7e^{i\pi},\ -\mathrm{ie}^{\frac{i2\pi}{3}},-\mathrm{ie}^{-\frac{i2\pi}{3}}$ $\Rightarrow z=e^{-\frac{\mathrm{i}\pi}{2}}7e^{i\pi},\ e^{-\frac{\mathrm{i}\pi}{2}}\mathrm{e}^{\frac{i2\pi}{3}},e^{-\frac{\mathrm{i}\pi}{2}}\mathrm{e}^{-\frac{i2\pi}{3}}$ $\Rightarrow z=7e^{\frac{i\pi}{2}},\ \mathrm{e}^{\frac{i\pi}{6}},\mathrm{e}^{\frac{i5\pi}{6}}$	
4(i)	$x-1=3\tan\theta$ $\dfrac{\mathrm{d}x}{\mathrm{d}\theta}=3\sec^2\theta$ $\displaystyle\int\frac{1}{\sqrt{x^2-2x+10}}\,\mathrm{d}x=\int\frac{1}{\sqrt{(x-1)^2+3^2}}\,\mathrm{d}x$ $\displaystyle=\int\frac{1}{\sqrt{(3\tan\theta)^2+3^2}}\cdot3\sec^2\theta\ \mathrm{d}\theta$ $\displaystyle=\int\frac{1}{3\sec\theta}\cdot3\sec^2\theta\ \mathrm{d}\theta$ $\displaystyle=\int\sec\theta\ \mathrm{d}\theta$ $=\ln\lvert\sec\theta+\tan\theta\rvert+C$ $=\ln\left\lvert\dfrac{\sqrt{x^2-2x+10}}{3}+\dfrac{x-1}{3}\right\rvert+C$	Many candidates omitted the square root in the denominator when doing the substitution or applying the trigo identity. Apply **$\tan^2\theta+1=\sec^2\theta$** Remember the modulus sign, arbitrary constant and rewrite expression in terms of x. Use the right angle triangle to give answer in terms of x. $x-1=3\tan\theta\Rightarrow\tan\theta=\dfrac{x-1}{3}$ 

Qn		Remarks
4(ii)	$x+3=\frac{1}{2}(2x-2)+4$ $\int \frac{x+3}{\sqrt{x^2-2x+10}}\,dx$ $=\int \frac{\frac{1}{2}(2x-2)+4}{\sqrt{x^2-2x+10}}\,dx$ $=\frac{1}{2}\int \frac{2x-2}{\sqrt{x^2-2x+10}}\,dx+\int \frac{4}{\sqrt{(x-1)^2+3^2}}\,dx$ $=\frac{1}{2}\frac{\sqrt{x^2-2x+10}}{\frac{1}{2}}+4\int \frac{1}{\sqrt{(x-1)^2+3^2}}\,dx$ $=\sqrt{x^2-2x+10}+4\ln\left\vert\frac{\sqrt{x^2-2x+10}}{3}+\frac{x-1}{3}\right\vert+C$ Standard form integral $\int \underbrace{(2x-2)}_{f'(x)}\underbrace{(x^2-2x+10)}_{f(x)}^{-\frac{1}{2}}dx=\frac{(x^2-2x+10)^{-\frac{1}{2}+1}}{-\frac{1}{2}+1}+C$ Many students erroneously applied the formula in MF26 when the form of the integral is not the same. Ans in (i)	
5(i)	$f(r)-f(r+1)=\frac{\sqrt{r}}{2\sqrt{r}+1}-\frac{\sqrt{r+1}}{2\sqrt{r+1}+1}$ $=\frac{\left(2\sqrt{r}\sqrt{r+1}+\sqrt{r}\right)-\left(2\sqrt{r+1}\sqrt{r}+\sqrt{r+1}\right)}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}$ $=\frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}$ $\sum_{r=1}^{n}\frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}=\sum_{r=1}^{n}\left[f(r)-f(r+1)\right]$ $=f(1)-f(2)$ $+f(2)-f(3)$ $+\ldots$ $+f(n)-f(n+1)$ $=f(1)-f(n+1)$ $=\frac{1}{3}-\frac{\sqrt{n+1}}{2\sqrt{n+1}+1}$	

Qn		Remarks
5(ii)	$\displaystyle\sum_{r=1}^{\infty}\frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}$ $\displaystyle=\lim_{n\to\infty}\left(\frac{1}{3}-\frac{\sqrt{n+1}}{2\sqrt{n+1}+1}\right)$ $\displaystyle=\lim_{n\to\infty}\left(\frac{1}{3}-\frac{1}{2+\frac{1}{\sqrt{n+1}}}\right)$ $\displaystyle=-\frac{1}{6}$	Most students have the misconception that a fraction must go to 0 when the denominator goes to infinity. However, note that the numerator goes to infinity as well, thus the expression is indeterminate until further manipulation is done to get a clearer picture.
5(iii)	$\displaystyle\sum_{r=1}^{n}\frac{\sqrt{r+1}-\sqrt{r+2}}{\left(2\sqrt{r+1}+1\right)\left(2\sqrt{r+2}+1\right)}=\sum_{r=2}^{n+1}\frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}$ $\displaystyle=\sum_{r=1}^{n+1}\frac{\sqrt{r}-\sqrt{r+1}}{\left(2\sqrt{r}+1\right)\left(2\sqrt{r+1}+1\right)}-\frac{1-\sqrt{2}}{3(2\sqrt{2}+1)}$ $\displaystyle=\frac{1}{3}-\frac{\sqrt{n+2}}{2\sqrt{n+2}+1}-\frac{1-\sqrt{2}}{3(2\sqrt{2}+1)}$ $\displaystyle=\frac{\sqrt{2}}{2\sqrt{2}+1}-\frac{\sqrt{n+2}}{2\sqrt{n+2}+1}$ Need $\displaystyle\frac{\sqrt{2}}{2\sqrt{2}+1}-\frac{\sqrt{n+2}}{2\sqrt{n+2}+1}<-0.1$ <table><tr><td>n</td><td>$\frac{\sqrt{2}}{2\sqrt{2}+1}-\frac{\sqrt{n+2}}{2\sqrt{n+2}+1}$</td></tr><tr><td>56</td><td>-0.099797</td></tr><tr><td>57</td><td>-0.100043</td></tr></table> Using GC, least n = 57	One may alternatively input the expression as a summation into the GC, but the calculation of the sum for each n takes much longer. Some students were not careful with the number of decimal places, thus unable to compare the value with -0.1.

Qn		Remarks
6(i)	$y=1+\dfrac{2x+p}{x^2+x-6}$ $\dfrac{dy}{dx}=\dfrac{2\left(x^2+x-6\right)-(2x+p)(2x+1)}{\left(x^2+x-6\right)^2}$ $\dfrac{dy}{dx}=0\Rightarrow 2\left(x^2+x-6\right)-(2x+p)(2x+1)=0$ $2x^2+2px+12+p=0$ $4p^2-4(2)(12+p)>0$ $p^2-2p-24>0$ $(p+4)(p-6)>0$ $p<-4$ or $p>6$	Learn to use quotient rule.
6(ii)		Write down all the coordinates including the stationary points (in this case) as stated in the question.
6(iii)	$1+\sqrt{12-x^2}\geq\dfrac{2x+7}{(2-x)(x+3)}$ $1+\dfrac{2x+7}{(x-2)(x+3)}\geq-\sqrt{12-x^2}$ Sketch $y=-\sqrt{12-x^2}$ (as above)	Sketch $y=-\sqrt{12-x^2}$ on the same diagram with the end points touching the x-axis. Label both graphs clearly using 2 different colours e.g. blue, black or pencil.

Qn		Remarks
	$-2\sqrt{3} \le x < -3$ OR $-2.92 \le x \le 1.46$ OR $2 < x \le 2\sqrt{3}$	
7(i)	 Every horizontal line $y = k$ cuts the graph at most once. This implies g is one-one. Therefore g^{-1} exists $g^{-1} : x \mapsto \frac{1}{x} - 3, x \in \mathbb{R},\ x \neq 0$	There is a need to draw the graph to show that any horizontal line will cut the graph at most once. To show that function is 1-1, there is a need to have a general equation, $y = k$. This is different from "at only one point", as the line $y = 0$ does not cut the graph.
7(ii)	$\{x \in \mathbb{R} \mid x \neq 0\}$	Question asked for "solution set" therefore answer must be written in sets.
7(iii)	$R_{g^{-1}} = D_g = \mathbb{R} \setminus \{-3\}, D_f = \mathbb{R}^-$. Since $R_{g^{-1}} \not\subset D_f$, fg^{-1} does not exist.	
7(iv)	$k = -3$	
7(v)(a)	 $R_{fh^{-1}} = \left(0, e^{-9}\right)$	

Qn		Remarks
7(iv) (b)	$\mathrm{h}(x) = \mathrm{h}^{-1}(x)$ $\mathrm{h}(x) = x$ $\frac{1}{x+3} = x$ $x^2 + 3x - 1 = 0$ Since $x < -3$, $x = -3.30$ (3sf) Need to reject the other root which does not lie in to range $x < -3$.	
8(i)	$(x-1)^2 = (\mathrm{e}^t + \mathrm{e}^{-t})^2 = \mathrm{e}^{2t} + 2 + \mathrm{e}^{-2t}$ $(2y)^2 = (\mathrm{e}^t - \mathrm{e}^{-t})^2 = \mathrm{e}^{2t} - 2 + \mathrm{e}^{-2t}$ Hence $(x-1)^2 - (2y)^2 = 4$ $\frac{(x-1)^2}{2^2} - y^2 = 1$ Alternative solution by students: $(x-1) + 2y = 2\mathrm{e}^t$ ___(1) $(x-1) - 2y = 2\mathrm{e}^{-t}$ ___(2) (1)×(2): $(x-1)^2 - (2y)^2 = 4\mathrm{e}^t\mathrm{e}^{-t} = 4$	
8(ii)		Many students simply drew the whole hyperbola given by the Cartesian equation in **(i)** without taking into account the original parametric equations. Note that one needs to check the range of values the x and y coordinates can take. For all $t \in \mathbb{R}$, $x = 1 + \mathrm{e}^t + \mathrm{e}^{-t} > 0$. Just key in parametric equations in GC.

Qn | **Remarks**

8(iii)

When $x=3$, $3=1+\mathrm{e}^{t}+\mathrm{e}^{-t}$

$$\mathrm{e}^{t}+\mathrm{e}^{-t}=2$$

$$t=0$$

When $x=1+\mathrm{e}+\mathrm{e}^{-1}$, $t=\pm 1$ ($t=1:\ y>0,\ t=-1:\ y<0$)

$$x=1+\mathrm{e}^{t}+\mathrm{e}^{-t}$$

$$\frac{\mathrm{d}x}{\mathrm{d}t}=\mathrm{e}^{t}-\mathrm{e}^{-t}$$

By symmetry

Note that the integral $\int y\mathrm{d}x$ refers to the area <u>either</u> above the x-axis (if $y>0$) <u>or</u> below the x-axis (if $y<0$)

Area of required region $=2\int_{3}^{1+\mathrm{e}+\mathrm{e}^{-1}} y\mathrm{d}x$

$$=2\int_{0}^{1}\frac{\mathrm{e}^{t}-\mathrm{e}^{-t}}{2}\left(\mathrm{e}^{t}-\mathrm{e}^{-t}\right)\mathrm{d}t$$

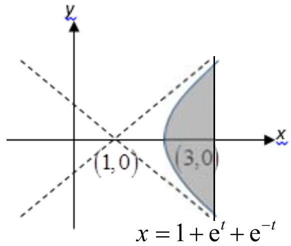
$$=\int_{0}^{1}\left(\mathrm{e}^{t}-\mathrm{e}^{-t}\right)^{2}\mathrm{d}t$$

$$=\int_{0}^{1}\left(\mathrm{e}^{2t}-2+\mathrm{e}^{-2t}\right)\mathrm{d}t$$

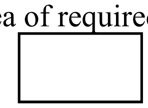
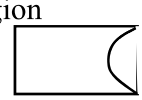
$$=\left[\frac{1}{2}\mathrm{e}^{2t}-2t-\frac{1}{2}\mathrm{e}^{-2t}\right]_{0}^{1}$$

$$=\left[\frac{1}{2}\mathrm{e}^{2}-2-\frac{1}{2}\mathrm{e}^{-2}\right]-0$$

$$=\frac{1}{2}\left(\mathrm{e}^{2}-\mathrm{e}^{-2}\right)-2$$



<u>Alternatively (more tedious)</u>:
Area of required region

=  − 

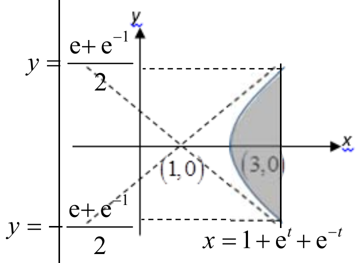
$$=\left(1+e+e^{-1}\right)\left(\frac{e-e^{-1}}{2}-\left(-\frac{e-e^{-1}}{2}\right)\right)-\int_{-\frac{\mathrm{e}-\mathrm{e}^{-1}}{2}}^{\frac{\mathrm{e}-\mathrm{e}^{-1}}{2}} x\ \mathrm{d}y$$

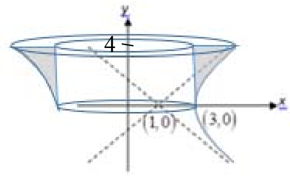
$$=\left(1+e+e^{-1}\right)\left(e-e^{-1}\right)-2\int_{0}^{\frac{\mathrm{e}-\mathrm{e}^{-1}}{2}} x\ \mathrm{d}y$$

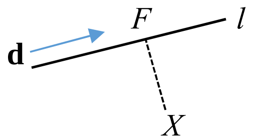
$$=\left(1+e+e^{-1}\right)\left(e-e^{-1}\right)-2\int_{0}^{1}\left(1+\mathrm{e}^{t}+\mathrm{e}^{-t}\right)\frac{\mathrm{e}^{t}+\mathrm{e}^{-t}}{2}\mathrm{d}t$$

$$\vdots$$

$$=\frac{1}{2}\left(\mathrm{e}^{2}-\mathrm{e}^{-2}\right)-2$$



Qn		Remarks
8(iv)	$\frac{(x-1)^2}{2^2}-y^2=1$ $(x-1)^2=2^2(1+y^2)$ $x=1+2\sqrt{1+y^2}\quad \text{since } x>1$ Volume $=\pi\int_0^4 x^2 \mathrm{d}y-\pi(3^2)(4)$ $=\pi\int_0^4\left(1+2\sqrt{1+y^2}\right)^2 \mathrm{d}y-36\pi$ $=335 \text{ units}^3 \quad (3 \text{ s.f.})$	 Note that finding volume of revolution when the curve is defined parametrically is not in syllabus. Students can use the parametric equations to find volume but are not expected to do so. You should just use the Cartesian equation.
9(i)	$\overrightarrow{OP}=\begin{pmatrix}\lambda-\mu\\1+2\mu\\2-3\lambda\end{pmatrix}=\begin{pmatrix}0\\1\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-3\end{pmatrix}+\mu\begin{pmatrix}-1\\2\\0\end{pmatrix}$ Locus of P is the plane with equation $\mathbf{r}=\begin{pmatrix}0\\1\\2\end{pmatrix}+\lambda\begin{pmatrix}1\\0\\-3\end{pmatrix}+\mu\begin{pmatrix}-1\\2\\0\end{pmatrix},\quad \lambda,\mu\in\mathbb{R}$	Students are expected to identify that the locus of P is a **plane** and to give the **correct equation** (in any acceptable form)
9(ii)	Normal of the locus of P (plane), $\begin{pmatrix}1\\0\\-3\end{pmatrix}\times\begin{pmatrix}-1\\2\\0\end{pmatrix}=\begin{pmatrix}6\\3\\2\end{pmatrix}$ $\begin{pmatrix}6\\3\\2\end{pmatrix}\cdot\begin{pmatrix}2\\-2\\-3\end{pmatrix}=0$ Hence the line l and the plane are parallel. Equation of the plane, $\mathbf{r}\cdot\begin{pmatrix}6\\3\\2\end{pmatrix}=\begin{pmatrix}0\\1\\2\end{pmatrix}\cdot\begin{pmatrix}6\\3\\2\end{pmatrix}=7$ $\begin{pmatrix}1\\1\\0\end{pmatrix}\cdot\begin{pmatrix}6\\2\\3\end{pmatrix}=8\neq 7$ Hence l is parallel to the plane and does not lie in the plane. Hence points P and Q will never meet. [Note that it is not sufficient just to show that l is parallel to the plane as it may actually lie on it. One must still need to show that there is a point on l that is not on the plane] Alternatively, one can check that	Many students made calculation error in the cross product. It is strongly advised that students **check the correctness** by either using the GC or simply taking the dot product between the resulting vector and any one of the two vectors and verify that it is zero. For e.g. $\begin{pmatrix}-1\\2\\0\end{pmatrix}\bullet\begin{pmatrix}6\\3\\2\end{pmatrix}=-6+6=0$ The marker pointed this out in the MY exam but apparently it has fallen on deaf ears

Qn		**Remarks**
	$\left[\begin{pmatrix}1\\1\\0\end{pmatrix}+t\begin{pmatrix}2\\-2\\-3\end{pmatrix}\right]\bullet\begin{pmatrix}6\\3\\2\end{pmatrix}\neq 7$ for all $t\in\mathbb{R}$.	
9(iii)	Shortest distance between P and Q is the distance between the line and the parallel plane. $\dfrac{\left\lvert\begin{pmatrix}1\\1\\0\end{pmatrix}\cdot\begin{pmatrix}6\\3\\2\end{pmatrix}-7\right\rvert}{\left\lvert\begin{pmatrix}6\\3\\2\end{pmatrix}\right\rvert}=\dfrac{2}{7}$	Many students used the wrong vector in this computation. Simply take a point on *l*, say (1,1,0) and compute its distance from the plane
9(iv)	Lines *l* and *m* are non-parallel. Hence $k\neq 1$. If the two lines intersect, $\begin{pmatrix}1\\k\\0\end{pmatrix}+s\begin{pmatrix}2\\-2\\-3k\end{pmatrix}=\begin{pmatrix}1\\1\\0\end{pmatrix}+t\begin{pmatrix}2\\-2\\-3\end{pmatrix}$ for some $s,t\in\mathbb{R}$ $1+2s=1+2t$ ___(1) $k-2s=1-2t$ ___(2) $-3sk=-3t$ ___(3) From (1), $s=t$ Substituting $s=t$ in (2), $k=1$ $s=t$ and $k=1$ satisfies (3) Thus for the system of linear equations to be inconsistent, $k\neq 1$. Hence lines *l* and *m* are skew when $k\neq 1$.	Students must understand that if two lines are skew, then they are: ▪ Non-parallel ▪ Non-intersecting Students are expected to justify that $k\neq 1$ is the condition on *k* that satisfy the above two requirements
9(v)	Let *F* be the foot of perpendicular from *X* to line *l*. $\overrightarrow{OF}=\begin{pmatrix}1\\1\\0\end{pmatrix}+t\begin{pmatrix}2\\-2\\-3\end{pmatrix}$ for some $t\in\mathbb{R}$ $\overrightarrow{XF}=\begin{pmatrix}1\\2\\5\end{pmatrix}+t\begin{pmatrix}2\\-2\\-3\end{pmatrix}$ $\left[\begin{pmatrix}1\\2\\5\end{pmatrix}+t\begin{pmatrix}2\\-2\\-3\end{pmatrix}\right]\cdot\begin{pmatrix}2\\-2\\-3\end{pmatrix}=0$ $-17+17t=0$ $t=1$ $\overrightarrow{OF}=\begin{pmatrix}1\\1\\0\end{pmatrix}+\begin{pmatrix}2\\-2\\-3\end{pmatrix}=\begin{pmatrix}3\\-1\\-3\end{pmatrix}$ $F\equiv(3,-1,-3)$	Students must understand that *XF* and <u>not *OF*</u> is perpendicular to *l*.  Therefore $\overrightarrow{XF}\bullet\mathbf{d}=0$

Qn		Remarks
10	Let S be the distance between the front of the car and the train at time t. $s=\sqrt{x^2+30^2}$ and $x^2=(40-30t)^2+(20t)^2$ $s=\sqrt{(40-30t)^2+(20t)^2+30^2}=\sqrt{1300t^2-2400t+2500}$	
10(i)	$\frac{ds}{dt}=\frac{1}{2}(1300t^2-2400t+2500)^{-\frac{1}{2}}(2600t-2400)$ When $t=1, \frac{ds}{dt}=2.67$	Generally ok. There are still handful of students not differentiating the expression given.
10(ii)	At stationary point $\frac{ds}{dt}=0$ $\frac{1}{2}(1300t^2-2400t+2500)^{-\frac{1}{2}}(2600t-2400)=0$ $2600t-2400=0 \Rightarrow t=\frac{12}{13}$ $\frac{d^2s}{dt^2}=\frac{1}{2}(1300t^2-2400t+2500)^{-\frac{1}{2}}(2600)$ $+(-\frac{1}{2})\frac{1}{2}(1300t^2-2400t+2500)^{-\frac{3}{2}}(2600t-2400)$ When $t=\frac{12}{13}, \frac{d^2s}{dt^2}>0$ $s=\sqrt{1300(\frac{12}{13})^2-2400(\frac{12}{13})+2500}=$ 19.884=19.9	
10(iii)	Let the angle of elevation be θ $\sin\theta=\frac{30}{\sqrt{1300t^2-2400t+2500}}$ $\cos\theta\frac{d\theta}{dt}=-\frac{1}{2}(30)(1300t^2-2400t+2500)^{-\frac{3}{2}}(2600t-2400)$ When $t=1$, $\cos\theta=\frac{\sqrt{(40-30)^2+20^2}}{\sqrt{1300-2400+2500}}=\sqrt{\frac{500}{1400}}$ $\therefore\frac{d\theta}{dt}=-\frac{1}{2}(30)(1300-2400+2500)^{-\frac{3}{2}}\left(200\right)\div\sqrt{\frac{500}{1400}}=-0.0958$ rad/s (or -5.5°/s)	Most of the students are not able to form the trigo equation.

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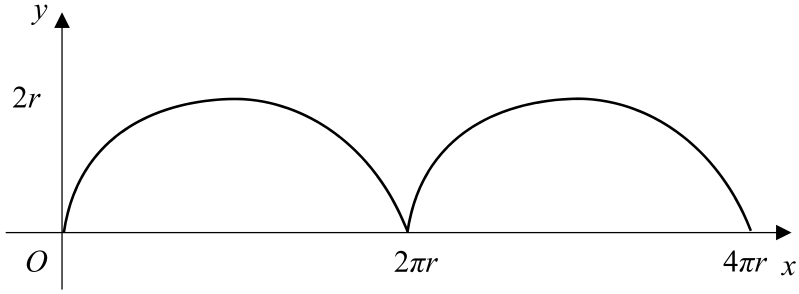
Qn		Remarks
11(i)		Candidates often forget to indicate the axial intercepts. For those whom had written down the intercepts, many simply wrote 2π and 4π without the r
11(ii)	$\frac{\mathrm{d}x}{\mathrm{d}\theta}=r(1-\cos\theta)$, $\frac{\mathrm{d}y}{\mathrm{d}\theta}=r\sin\theta$ $\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\sin\theta}{1-\cos\theta}$ $=\frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\sin^2\frac{\theta}{2}}$ Use do formula Use trigo identity $=\frac{\cos\frac{\theta}{2}}{\sin\frac{\theta}{2}}=\cot\frac{\theta}{2}$	Some candidates did not make use of double angle formula and end up using a long and tedious method to arrive at the given answer
11(iii)	$\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2=\cot^2\frac{\theta}{2}=\mathrm{cosec}^2\frac{\theta}{2}-1$ $=\frac{1}{\sin^2\frac{\theta}{2}}-1$ Use double angle formula $=\frac{2}{1-\cos\theta}-1=\frac{2r}{r(1-\cos\theta)}-1$ Thus $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2=\frac{2r}{y}-1$	Badly done with poor presentation. You are required to prove that $\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2=\frac{2r}{y}-1$. Many candidates started by stating this equation and proceed to prove. This is not true! The key is to apply double angle formula.
11(iv)	$Area=\int_0^{2\pi r} y\,dx$ $\mathrm{d}x=\frac{\mathrm{d}x}{\mathrm{d}\theta}\mathrm{d}\theta=r(1-\cos\theta)\mathrm{d}\theta$ $=\int_0^{2\pi} r(1-\cos\theta)\cdot r(1-\cos\theta)\,d\theta$ $=r^2\int_0^{2\pi}(1-2\cos\theta+\cos^2\theta)\,d\theta$ Use double angle formula $=r^2\int_0^{2\pi}(\frac{3}{2}-2\cos\theta+\frac{1}{2}\cos 2\theta)\,d\theta$ $=r^2\left[\frac{3}{2}\theta-2\sin\theta+\frac{1}{4}\sin 2\theta\right]_0^{2\pi}$ $=3\pi r^2$	Many candidates are unable to write out the area expression in parametric form correctly. A handful of students did not integrate the expression correctly.